Derick Amalraj

Location: Toronto, ON **Email:** derick.amalraj.overview@gmail.com

SUMMARY

Senior Software Development Engineer specializing in firmware validation infrastructure, continuous integration and delivery (CI/CD) systems, automated quality engineering, and test automation. Experienced in building scalable validation platforms, improving reliability through automation, and developing engineering processes that enable large cross-functional teams to deliver high-quality software and firmware at scale. Proven track record in infrastructure optimization, release-readiness reporting, and organization-wide quality initiatives.

SKILLS

- **Engineering:** Firmware Validation, Infrastructure Management, DevOps, Systems Integration Testing, Scalability, Performance Optimization, Root Cause Analysis, Reliability, Full-Stack Development, Machine Learning (Logistic Regression, Principal Component Analysis, Model Selection, Neural Networks, Monte Carlo Methods), API Design, RESTful Web Services, Microservices, Testing & Code Coverage, Automation, Code Reviews, Requirements Documentation
- **Infrastructure:** GitHub Actions, Jenkins, Kubernetes, Docker, Terraform, Helm, Elasticsearch, Proxmox, Virtualization, Cloud Infrastructure, Backend Services
- **Tools & Programming:** C/C++, CUDA/ROCm, Python, Groovy, Java, MATLAB, Haskell, VBA, JavaScript, TypeScript, LaTeX, VirtualBox, Jira, MongoDB, SQL, GraphQL, Unix Shell, VS Code
- **Leadership:** Technical Leadership, Stakeholder Communication, Quality Governance, Strategic Planning & Execution, Project Management, Process Improvement, Cross-functional Collaboration

RELEVANT EXPERIENCE

Senior Software Development Engineer, Advanced Micro Devices (AMD) **July 2026 - Present**

- Owned firmware quality infrastructure and automated validation workflows across multiple firmware components and hardware programs, providing readiness, reliability, and regression visibility to engineering leadership.
- Presented firmware validation strategy, CI process improvements, and quality initiatives to engineering teams and leadership, driving adoption of scalable support and quality-management practices.
- Developed and evolved automated firmware quality reporting systems summarizing 30–45+ validation jobs per execution, improving regression visibility, infrastructure diagnostics, and release-readiness assessment for engineering and leadership stakeholders.
- Implemented program-level health tracking, dashboards, KPI reporting, and validation analytics used to monitor test stability, failure trends, validation coverage, and CI reliability across server, client, and accelerator firmware programs.
- Coordinated organization-wide rollout of validation requirements and quality gates across firmware repositories, improving visibility into new checks, infrastructure changes, and validation expectations.
- Designed and operated a centralized firmware quality communication model supporting over 450 engineers and managers, improving visibility into validation requirements, infrastructure incidents, and automated quality gate rollouts.
- Reduced simulation execution time by over 70% (2 hours $\rightarrow$ 15 minutes) through resource scheduling improvements, test restructuring, and virtual machine capacity optimization.
- Expanded pre-merge and nightly validation coverage across BIOS, platform, power-management, and security firmware, enabling earlier defect detection while maintaining validation stability and throughput.
- Built and maintained automated validation environments, virtualized test infrastructure, and custom operating system images used to support scalable firmware testing, deployment, debugging, and failure recovery workflows.
- Developed and automated simulation-based validation workflows using Python and shell scripting to verify firmware quality across multiple hardware platforms.
- Refactored and redesigned the internal CI infrastructure with a focus on modularity and reusability to consolidate previously incompatible code and accommodate the growing testing needs of the CI team and inbound requests.
- Designed and introduced scalable CI communication, support, and onboarding processes that improved engineering visibility, supportability, and cross-team coordination.
- Received 3 AMD Spotlight awards for infrastructure management and post-silicon/onboard testing.

ADDITIONAL EXPERIENCE

- **Software Development Engineer 2, Advanced Micro Devices (AMD)** **April 2024 - June 2026**
- **Teaching Assistant, University of Toronto** **January 2023 - December 2023**
- **Software Engineer Intern, Advanced Micro Devices (AMD)** **May 2021 - August 2022**

EDUCATION

- **Master of Science in Applied & Computational Mathematics, University of Washington** **Spring 2026**
- **Bachelor of Science in Computer Science & Mathematics with Honours, University of Toronto** **Summer 2023**

SELECT PROJECTS

Path-Tracer

- Engineered a path tracer in C that enables users to input and transform specific data points into light path representations.
- Integrated ray tracing algorithms and various techniques for analyzing shadows, reflections, and global illumination, such as texture mapping, anti-aliasing, triangle meshes, and nonlinear transformations.
- Cut rendering time by 30% while preserving image quality by integrating Monte Carlo sampling and bounding volume hierarchies to accelerate ray tracing.